Riemannian Manifolds

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Abstract

Contents

1 Riemannian Metrics

1.1 The Riemannian Metric

Definition 1.1.1 (Riemannian Metric)
Let M be a smooth manifold. A Riemannian metric on M is a smooth section

$$g \in \mathcal{C}^\infty(M, (TM \otimes TM)^*)$$

such that $g_p : T_p(M) \times T_p(M) \to \mathbb{R}$ is an inner product for all $p \in M$.

Equivalently, a Riemannian metric is an assignment $p \mapsto g_p$ such that for any smooth sections $s, t \in \Gamma(M, TM)$, the map $p \mapsto g_p(s(p), t(p))$ is smooth.

Definition 1.1.2 (Riemannian Manifold) A Riemannian manifold (M, g) is a manifold M together with a Riemannian metric g on M.

To describe the Riemannian metric in local coordinates, we need some simple algebra. Let V be a finite dimensional vector space over a field k. Let $\{e_1, \dots, e_n\}$ be a basis for V. Let $\{e^1, \dots, e^n\}$ be the dual basis for V^*. Then we may establish a canonical isomorphism

$$(V \otimes_k V)^* \cong V^* \otimes_k V^*$$

given by $f(-,-) \mapsto \sum_{i,j=1}^n f(e_i \otimes e_j) e^i \otimes e^j$. Its converse is given by $\varphi \otimes \psi \mapsto ((v \otimes w) \mapsto \varphi(v)\psi(w))$.

Then isomorphism carries over to vector bundles, and so we have an isomorphism $(TM \otimes TM)^* \cong TM^* \otimes TM^*$. This is how the Riemannian metric is a $(0, 2)$-tensor field.

Proposition 1.1.3
Let (M, g) be a Riemannian manifold. Let $p \in M$. Let $(U, x^1, \dots, x^n)$ be a chart at p. Then g_p as an element of $T_p(M)^* \otimes T_p(M)^*$ is given locally on the chart by

$$g_p = \sum_{1 \le i,j \le n} g_{i,j}(p) \; (dx^i)|_p \otimes (dx^j)_p$$

where $g_{i,j} : U \to \mathbb{R}$ are smooth functions such that $g_{i,j}(p) = g_p \left(\left.\frac{\partial}{\partial x^i}\right|_p, \left.\frac{\partial}{\partial x^j}\right|_p \right)$.

Let $v, w \in T_p(M)$ be tangent vectors. Then locally on the chart $(U, x^1, \dots, x^n)$, we may compute the Riemannian metric by

$$g_p(v, w) = \sum_{1 \le i,j \le n} g_{i,j}(p)(dx^i)|_p(v)(dx^j)|_p(w)$$

Theorem 1.1.4
Every smooth manifold admits a Riemannian metric and hence is a Riemannian manifold.

Definition 1.1.5 (Isometries) Let (M, g) and (N, h) be two Riemannian manifolds. We say that (M, g) and (N, h) are isometric if there exists a diffeomorphisms $f : M \to N$ such that

$$h \circ f = g$$

In this case f is said to be an isometry.

Theorem 1.1.6 ((Nash))
Let (M, g) be a Riemannian manifold of dimension n. Then there exists an isometric embedding $f : M \to \mathbb{R}^{(n+2)(n+3)/2}$ with the standard Euclidean metric.

Definition 1.1.7 (Local Isometries) Let (M,g) and (N,h) be two Riemannian manifolds. We say that they are locally isometric if for all $p\in M$, there exists an open neighbourhood $U\subseteq M$ of p and $V\subseteq N$ open and an isometry $f:U\to V$.

Definition 1.1.8 (Flat Manifolds) Let (M,g) be a Riemannian manifold. We say that (M,g) is flat if it is locally isometric to $\mathbb{R}^n$ with the standard metric.

In general, not every Riemannian manifold is flat. This can be shown once we discuss curvatures and torsions. However, this is true when $n=1$.

Lemma 1.1.9 Every 1 dimensional Riemannian manifold is flat.

1.2 Lengths and Angles

Definition 1.2.1 (Length of a Tangent Vector) Let (M,g) be a Riemannian manifold. Let $v\in T_p(M)$ be a tangent vector for $p\in M$. Define the length of v to be

$$|v|_g=\sqrt{g_p(v,v)}$$

Definition 1.2.2 (Angle between two Tangent Vectors) Let (M,g) be a Riemannian manifold. Let $p\in M$. For $v,w\in T_pM$ two tangent vectors, define the angle between v and w to be the unique $\theta\in[0,\pi]$ such that

$$\cos(\theta)=\frac{g_p(v,w)}{|v|_g|w|_g}$$

Definition 1.2.3 (Orthogonal Tangent Vectors) Let (M,g) be a Riemannian manifold. Let $p\in M$. We say that two tangent vectors $v,w\in T_pM$ are orthogonal if

$$g_p(v,w)=0$$

1.3 The Musical Isomorphisms

Definition 1.3.1 (The Flat Map) Let (M,g) be a Riemannian manifold. Let $p\in M$. For each $X\in T_pM$, define the flat map

$$\flat:T_pM\to T_p^*M$$

by sending $X\in T_pM$ to the map $X^\flat:T_pM\to\mathbb{R}$ by $X^\flat(Y)=g_p(X,Y)$.

Theorem 1.3.2 (The Musical Isomorphism) Let (M,g) be a Riemannian manifold. Let $p\in M$. Then the flat map

$$\flat:T_pM\to T_p^*M$$

is an isomorphism.

Definition 1.3.3 (The Sharp Map) Let (M,g) be a Riemannian manifold. Let $p\in M$. Define the sharp map

$$\#:T_p^*M\to T_pM$$

to be the inverse of the flat map.

Example 1.3.4
Let (M,g) be a Riemannian manifold. Let $p\in U$. Let $(U,x^1,\dots,x^n)$ be a chart at p. Let g be given

locally at the chart by
$$g_p = \sum_{1\leq i,j\leq n} g_{i,j}(p)(dx^i)_p \otimes (dx^j)_p$$
Then the following are true.
- The flat map is given locally by
$$\left(\frac{\partial}{\partial x^i}\bigg|_p\right)^\flat = \sum_{j=1}^n g_{i,j}(p)(dx^j)_p$$
- The sharp map is given locally by
$$((dx^i)_p)^\# = \sum_{j=1}^n g^{i,j}(p)\frac{\partial}{\partial x^j}\bigg|_p$$
where $(g^{i,j}(p))_{1\leq i,j\leq n} = ((g_{i,j}(p))_{1\leq i,j\leq n})^{-1}$.

1.4 The Induced Metric on Cotangent Bundles

Definition 1.4.1 (The Induced Metric on the Cotangent Bundle)
Let (M,g) be a Riemannian manifold. Define the induced metric on T^*M to be the section $g^{-1}\in\Gamma(M,(T^*M\otimes T^*M)^*)$ by
$$g^{-1}(\alpha,\beta) = g(\alpha^\#,\beta^\#)$$

Proposition 1.4.2
Let (M,g) be a Riemannian manifold. Let $(U,x^1,\dots,x^n)$ be a chart of M. Suppose that the components of g^{-1} on the local chart is given by $g^{i,j}$. Then we have
$$(g^{i,j}) = (g_{i,j})^{-1}$$

1.5 Riemannain Manifolds as a Metric Space

Definition 1.5.1 (Piecewise Regular Paths)
Let (M,g) be a Riemannian manifold. A piecewise regular path in M consists of smooth functions $\gamma_i:[a_i,b_i]\to M$ for $1\leq i\leq k$ such that the following are true.
- For $1\leq i\leq k-1$, we have $\gamma_i(b_i)=\gamma_{i+1}(a_i)$.
- For $1\leq i\leq k$, we have $|\gamma_i'(t)|\neq 0$ for all $t\in[a_i,b_i]$.

Definition 1.5.2 (Length of a Curve)
Let (M,g) be a Riemannian manifold. Let $\gamma:[a,b]\to M$ be a piecewise regular path in M. Define the length of γ to be
$$L(\gamma) = \int_a^b |\gamma'(t)|_g\,dt$$

Definition 1.5.3 (Riemannian Distance Function)
Let (M,g) be a Riemannian manifold. Define the Riemannian distance function of M to be the function $d_g: M\times M\to\mathbb{R}$ given by
$$d_g(p,q) = \inf\{L(\gamma)\mid \gamma: I\to M \text{ is a piecewise regular path }\}$$

Proposition 1.5.4
Let (M,g) be a Riemannian manifold. Then the following are true.
- d_g is a metric on M.
- The topology of M is given by the metric topology induced by d_g.

2 Introduction to Connections

2.1 Connections on Vector Bundles

Recall that for a smooth vector bundle $p : E \to M$, we denote the space of smooth sections on E by $\Gamma(E)$. Moreover, $\mathfrak{X}(M)$ is the space of smooth sections on the tangent bundle TM.

Definition 2.1.1 (Connections)
Let M be a smooth manifold. Let $p : E \to M$ be a smooth vector bundle. A connection on p is a map

$$\nabla : \mathfrak{X}(M) \times \mathcal{C}^\infty(E) \to \mathcal{C}^\infty(E)$$

where we denote $\nabla(V, T)$ by $\nabla_V(T)$, such that the following are true.

- $C^\infty(M)$-linearity in first variable: For each $T \in \mathcal{C}^\infty(E)$, the map $V \mapsto \nabla_V(T)$ is $C^\infty(M)$-linear. This means that
$$\nabla_{fV+hW}(T) = f\nabla_V(T) + g\nabla_W(T)$$
for $V, W \in \mathfrak{X}(M)$, $f, g \in C^\infty(M)$.
- $\mathbb{R}$-linearity in second variable: For each $V \in \mathfrak{X}(M)$, the map $T \mapsto \nabla_V(T)$ is $\mathbb{R}$-linear. This means that
$$\nabla_V(\lambda T + \mu S) = \lambda\nabla_V(T) + \mu\nabla_V(S)$$
- Product rule: The map ∇ satisfies the following product rule:
$$\nabla_V(fT) = V(f) \cdot T + f\nabla_V(T)$$

Proposition 2.1.2
Let M be a smooth manifold. Let $p : E \to M$ be a smooth vector bundle. Then E admits a connection.

Definition 2.1.3 (Set of Connections)
Let M be a smooth manifold. Let $p : E \to M$ be a smooth vector bundle. Define the set of connections of E to be

$$\mathcal{A}(E) = \{\nabla : \mathfrak{X}(M) \times \mathcal{C}^\infty(E) \to \mathcal{C}^\infty(E) \mid \nabla \text{ is a connection } \}$$

Proposition 2.1.4
Let M be a smooth manifold. Let $p : E \to M$ be a smooth vector bundle. A connection ∇ on E is equivalently an $\mathbb{R}$-linear map

$$\nabla : \mathcal{C}^\infty(E) \to \mathcal{C}^\infty(T^*M \otimes E) = \Omega^1(M) \otimes_{\mathcal{C}^\infty(M)} \mathcal{C}^\infty(E)$$

such that

$$\nabla(f \cdot s) = df \otimes s + f \cdot \nabla s$$

for all $f \in \mathcal{C}^\infty(M)$ and $s \in \mathcal{C}^\infty(E)$.

Definition 2.1.5 (The Connection Forms)
Let M be a smooth manifold. Let $p : E \to M$ be a smooth vector bundle. Let ∇ be a connection on E. Let $U \subseteq M$ be an open subset. Let $e_1, \dots, e_r$ be a local frame of E on U. Define the connection 1-forms of ∇ on U to be the unique 1-forms $\omega_j^i : TU \to \mathbb{R}$ such that

$$\nabla_X e_j = \sum_{i=1}^r \omega_j^i(X) e_j$$

for all $X \in \mathfrak{U}$.

Proposition 2.1.6
Let M be a smooth manifold. Let $p : E \to M$ be a smooth vector bundle. Let ∇ be a connection on E. Let $U \subseteq M$ be an open subset. Let $e, \overline{e}$ be local frames of E on U. Suppose that $g : U \to \mathrm{GL}(r, \mathbb{R})$ is a transition function such that $\overline{e} = ge$. Suppose that ω_j^i and $\overline{\omega}_j^i$ are the connection forms of ∇ with respect to e and $\overline{e}$. Then we have

$$\overline{\omega}_j^i = g^{-1}\omega g + g^{-1}dg$$

2.2 Directional Derivatives of a Vector Bundle

Definition 2.2.1 (Covariant Derivatives Along a Curve)
Let M be a smooth manifold. Let $p : E \to M$ be a smooth vector bundle. Let $\gamma : [0,1] \to M$ be a smooth curve. Let ∇ be a connection on E. Define the covariant derivative $D_t : \mathcal{C}^\infty(\gamma^*(E)) \to \mathcal{C}^\infty(\gamma^*(E))$ of γ to be the pullback of the connection ∇ on γ.

Proposition 2.2.2
Let M be a smooth manifold. Let $p : E \to M$ be a smooth vector bundle. Let $\gamma : [0,1] \to M$ be a smooth curve. Let ∇ be a connection on E. Then the covariant derivative is given in local charts $(U, \phi = (x^1, \dots, x^n))$ for $\gamma(t_0) \in M$ by the formula

$$(D_tV)_{\gamma(t_0)} = \sum_{i=1}^n \left(\frac{\partial a_i}{\partial x^i}\bigg|_{\gamma(t_0)} \frac{\partial}{\partial x^i}\bigg|_{\gamma(t_0)} + a_i(\gamma(t_0))\nabla\left(\gamma'(t_0), \frac{\partial}{\partial x^i}\bigg|_{\gamma(t_0)}\right)\right)$$

where V is given locally on the same chart by

$$V_{\gamma(t)} = \sum_{i=1}^n a_i(\gamma(t)) \frac{\partial}{\partial x^k}\bigg|_{\gamma(t)}$$

Definition 2.2.3 (Parallel Sections)
Let M be a smooth manifold. Let $p : E \to B$ be a smooth vector bundle. Let $\gamma : [0,1] \to M$ be a smooth curve. Let ∇ be a connection on E. Define the parallel sections of γ along ∇ to be $\ker(D_t)$.

In other words, a parallel section is a local section $s : \mathrm{im}(\gamma) \to E$ such that $D_t(s) = 0$.

Proposition 2.2.4
Let M be a smooth manifold. Let $p : E \to B$ be a smooth vector bundle. Let $\gamma : [0,1] \to M$ be a smooth curve. Let ∇ be a connection on E. Let $X \in E_{\gamma(0)}$. Then there exists a unique parallel section $s : [0,1] \to E$ such that $s(0) = X$.

Definition 2.2.5 (Parallel Transports)
Let M be a smooth manifold. Let $p : E \to B$ be a smooth vector bundle. Let $\gamma : [0,1] \to M$ be a smooth curve. Let ∇ be a connection on E. Define the parallel transport map

$$P_\gamma : E_{\gamma(0)} \to E_{\gamma(1)}$$

by $X \mapsto s_X(1)$ where $s_X : [0,1] \to E$ is the unique parallel section such that $s_X(0) = X$.

Lemma 2.2.6
Let M be a smooth manifold. Let $p : E \to B$ be a smooth vector bundle. Let $\gamma : [0,1] \to M$ be a smooth curve. Let ∇ be a connection on E. Then P_γ is a well defined linear map.

2.3 The Curvature Tensor

Definition 2.3.1 (The Curvature Tensor)
Let M be a smooth manifold. Let $p : E \to M$ be a smooth vector bundle. Let ∇ be a connection on p. Define the curvature of ∇ to be the smooth section $R(\nabla) \in \mathcal{C}^\infty(\text{End}(E) \otimes \Lambda^2 T^*M)$ given by

$$R(\nabla)(X,Y)(s) = [\nabla_X, \nabla_Y](s) - \nabla_{[X,Y]}(s)$$

for $X, Y \in \mathfrak{X}(M)$ and $s \in \mathcal{C}^\infty(E)$.

Notice that the definition makes sense: Appealing to the tensor characterization lemma, a section of $\mathcal{C}^\infty(\text{End}(E) \otimes \Lambda^2 T^*M)$ corresponds to a homomorphism $\Lambda^2 TM \to \mathcal{C}^\infty(\text{End}(E))$. Applying the lemma to $\mathcal{C}^\infty(\text{End}(E)) = \mathcal{C}^\infty(E^* \otimes E)$ again, we see that the curvature tensor corresponds to a map

$$\Lambda^2 TM \to \text{Hom}_{\mathcal{C}^\infty(M)}(\mathcal{C}^\infty(E), \mathcal{C}^\infty(E))$$

Definition 2.3.2 (The Curvature Form)
Let M be a smooth manifold. Let $p : E \to M$ be a smooth vector bundle. Let ∇ be a connection on E. Let $U \subseteq M$ be an open subset. Let $e_1, \dots, e_r$ be a local frame of E on U. Define the curvature forms of ∇ on U to be the unique two forms $\Omega^i_j \in \mathcal{C}^\infty(U, E)$ such that

$$R(\nabla)(X,Y)(e_j) = \sum_{i=1}^r \Omega^i_j(X,Y) e_j$$

for all $X, Y \in \mathfrak{U}$.

Theorem 2.3.3 (The Second Structural Equation)
Let M be a smooth manifold. Let $p : E \to M$ be a smooth vector bundle. Let ∇ be a connection on E. Let $U \subseteq M$ be an open subset. Let $e_1, \dots, e_r$ be a local frame of E on U. Then we have

$$\Omega^i_j = d\omega^i_j + \sum_{k=1}^r \omega^i_k \wedge \omega^k_j$$

Another way to think about it: The matrix $\Omega = (\Omega^i_j)$ is an element of $\Omega^2(U, \mathfrak{gl}(n, \mathbb{R}))$ (indeed locally $\text{End}(E)$ is isomorphic to $\mathfrak{gl}(n, \mathbb{R})$). Gluing along local patches give $\Omega = d\omega + \omega \wedge \omega$. One can prove that $R(X,Y) = \Omega(X,Y)$. Indeed, Ω comes from the coordinate description of R, so patching it up should recover R.

Proposition 2.3.4
Let M be a smooth manifold. Let $p : E \to M$ be a smooth vector bundle. Let ∇ be a connection on E. Let $U \subseteq M$ be an open subset. Let $e, \overline{e}$ be local frames of E on U. Suppose that $g : U \to \text{GL}(r, \mathbb{R})$ is a transition function such that $\overline{e} = ge$. Suppose that Ω^i_j and $\overline{\Omega}^i_j$ are the curvature forms of ∇ with respect to e and $\overline{e}$. Then we have

$$\overline{\Omega}^i_j = g^{-1} \Omega g$$

3 Connections on the Tangent Bundle

3.1 Affine Connections

Definition 3.1.1 (Affine Connections) Let M be a smooth manifold. An affine connection of M is a connection
$$\nabla : \mathfrak{X}(M) \times \mathfrak{X}(M) \to \mathfrak{X}(M)$$
on the tangent bundle TM.

The directional derivative is the canonical affine connection on $\mathbb{R}^n$. In fact, every smooth manifold has such a canonical connection that generalizes the directional derivative.

Definition 3.1.2 (The Torsion Tensor)

3.2 The Levi-Civita Connection

Definition 3.2.1 (The Lie Bracket on Smooth Vector Fields) Let M be a smooth manifold. Let $X, Y \in \mathfrak{X}(M)$ be smooth vector fields. Define the Lie bracket of X and Y to be the vector field $[X, Y]$ given by the formula
$$[X, Y]_p(f) = X(Y_p(f)) = Y(X_p(f))$$
for $p \in M$ and $f \in \mathcal{C}^\infty_{M,p}$.

Proposition 3.2.2 Let M be a smooth manifold. Let $X, Y \in \mathfrak{X}(M)$ be smooth vector fields. Let (U, ϕ) be a chart on M. Let X and Y be given locally on the chart by the formula
$$X = \sum_{k=1}^n a_k \frac{\partial}{\partial x^k} \quad \text{and} \quad Y = \sum_{k=1}^n b_k \frac{\partial}{\partial x^k}$$
for $a_k, b_k : U \to \mathbb{R}$ smooth functions. Then the Lie bracket of X and Y is given locally on the chart by the formula
$$[X, Y] = \sum_{k=1}^n (X(b_k) - Y(a_k)) \frac{\partial}{\partial x^k} = \sum_{i=1}^n \left(\sum_{j=1}^n \left(a_j \frac{\partial b_i}{\partial x^i} - b_i \frac{\partial a_j}{\partial x^j} \right) \right) \frac{\partial}{\partial x^i}$$

Definition 3.2.3 (Metric Connections) Let M be a smooth manifold. Let ∇ be an affine connection. We say that ∇ is a metric connection if the
$$\nabla(X, g(Y, Z)) = g(\nabla(X, Y), Z) + g(Y, \nabla(X, Z))$$
for all $X, Y, Z \in \mathfrak{X}(M)$.

Definition 3.2.4 (Levi-Civita Connections)
Let (M, g) be a Riemannian manifold. Let $\nabla^g : \mathfrak{X}(M) \times \mathfrak{X}(M) \to \mathfrak{X}(M)$ be an affine connection on M. We say that ∇^g is a Levi-Civita connection if the following are true.
- ∇^g is a metric connection.
- For any $X, Y \in \mathfrak{X}(M)$, we have
$$\nabla^g(X, Y) - \nabla^g(Y, X) = [X, Y]$$

Theorem 3.2.5 (Existence and Uniqueness of Levi-Civita Connections) Let (M, g) be a Riemannian manifold. Then M has a unique Levi-Civita connection.

Theorem 3.2.6 (The Decomposition Theorem)

Proposition 3.2.7
Let (M, g) be a Riemannian manifold. Then M is flat if and only if $R(\nabla^g) = 0$.

Recall: Flat iff locally isometric to $\mathbb{R}^n$.

3.3 Ricci Curvature

Definition 3.3.1 (Ricci Curvature)
Let (M, g) be a Riemannian manifold. Let $p \in M$. Define the Ricci curvature of M to be the map Ric : $TM \times TM \to \mathbb{R}$ given by

$$\text{Ric}(X, Y) = \text{tr}(Z \mapsto R(\nabla^g)(Z, X)(Y))$$

4 Geometry on Riemannian Manifolds

4.1 Geodesics on Riemannian Manifolds

Let M be a smooth manifold. Let $\gamma : I \to M$ be a smooth curve on M. Then we can compute its differential to obtain a tangent vector

$$\gamma'(t) \in T_{\gamma(t)}M$$

for each $t \in I$. This defines a vector field in $\mathfrak{X}(\gamma)$.

Definition 4.1.1 (Geodesics) Let M be a smooth manifold. Let ∇ be an affine connection on M. A geodesic on M is a curve $\gamma : I \to M$ such that

$$D_t(\gamma'(t)) = 0$$

where D_t is the associated covariant derivative of ∇ and γ.

We have seen geodesics for metric spaces, but these are slightly different. We require geodesics on manifolds to only minimize the distance locally.

4.2 Minimizing Curves

Definition 4.2.1 (Minimizing Curves)
Let (M, g) be a Riemannian manifold. Let $\gamma : [a, b] \to M$ be a smooth curve in M. We say that γ is minimizing if

$$L(\gamma) = d_g(\gamma(a), \gamma(b))$$

Definition 4.2.2 (Locally Minimizing Curves)
Let (M, g) be a Riemannian manifold. Let $\gamma : [a, b] \to M$ be a smooth curve in M. We say that γ is locally minimizing if for all $t \in [a, b]$, there exists a neighbourhood $[c, d] \subseteq [a, b]$ of t such that $\gamma|_{[c,d]}$ is a minimizing curve.

Proposition 4.2.3
Let (M, g) be a Riemannian manifold. Let $\gamma : [a, b] \to M$ be a smooth curve in M. Then the following are true.

- If γ is a minimizing curve, then γ is a geodesic.
- If γ is a geodesic, then γ is a locally minimizing curve.

4.3 Geodesically Complete Riemannian Manifolds

Definition 4.3.1 (Geodesically Complete)

Proposition 4.3.2
Geodesically complete iff exp map well defined

Theorem 4.3.3 (Hopf-Rinow Theorem)
Let (M, g) be a connected Riemannian manifold. Then the following are equivalent.

- M is geodeically complete.
- M is complete as a metric space.
- Every closed and bounded set in M is compact.

Proposition 4.3.4
Let (M, g) be a connected Riemannian manifold. Let $p, q \in M$. Then there exists a minimizing geodesic from p to q.

5 Gauss-Bonnet Theorem

5.1 The Second Fundamental Form

5.2 Gauss-Bonnet Theorem

Theorem 5.2.1 (The Gauss-Bonnet Formula) Let (M, g) be an oriented smooth 2-manifold. Let γ be a positively oriented curved polygon in M and let Ω be its interior. Then

$$\int_\Omega K\,dA + \int_\gamma \kappa_N\,ds + \sum_{i=1}^k \varepsilon_i = 2\pi$$

where
- K is the Gaussian curvature of g
- dA is the Riemannian volume form
- ε_i are the exterior angles of γ
- The second integral is taken with respect to arc length

Theorem 5.2.2 (Gauss-Bonnet Theorem) Let (M, g) be an smooth compact 2-dimensional Riemannian manifold. Let K be the Gaussian curvature of M and let k_g be the geodesic curvature of ∂M. Then

$$\int_M K\,dA + \int_{\partial M} k_g\,ds = 2\pi\chi(M)$$

Corollary 5.2.3 Let (M, g) be an smooth compact 2-dimensional Riemannian manifold without boundary. Let K be the Gaussian curvature of M. Then

$$\int_M K\,dA = 2\pi\chi(M)$$

6 Calculus on Manifolds

6.1 Integrating Functions on Riemannian Manifolds

Definition 6.1.1 (The Riemannian Volume Form)
Let (M,g) be an orientable Riemannian manifold. Define the volume form dV of M to be given locally at $p\in M$ on the chart $(U,x^1,\dots,x^n)$ by

$$(dV)_p=\sqrt{\det(g_{i,j}(p))}(dx^1)_p\wedge\cdots\wedge(dx^n)_p$$

where g is given locally at the chart by $g_p=\sum_{1\leq i,j\leq n}g_{i,j}(p)(dx^i)_p\otimes(dx^j)_p$.

Let (M,g) be a compact orientable Riemannian manifold. Let $f:M\to\mathbb{R}$ be a compactly supported smooth function. Then we may define the integral of f over M to be the value $\int_M f dV$.

Definition 6.1.2 (The Volume of a Compact Orientable Manifold)
Let (M,g) be a compact orientable Riemannian manifold. Define the volume of M to be

$$\text{Vol}(M)=\int_M 1dV$$

Definition 6.1.3 (The Induced Volume Form at the Boundary)
Let (M,g) be an orientable Riemannian manifold with boundary. Define the induced volume form on ∂M to be

$$dS=\iota^*dV$$

Lemma 6.1.4
Let (M,g) be an orientable Riemannian manifold with boundary. Let (U,ϕ) be a boundary chart of M. Then locally on U, the induced volume form is given by

$$dS=\sqrt{\det(g_{ij}|_{\partial M})}dx^1\wedge\cdots\wedge dx^n$$

6.2 Differential Operators

Definition 6.2.1 (The Gradient of a Function)
Let (M,g) be a Riemannian manifold. Let $f:M\to\mathbb{R}$ be a smooth function. Define the gradient of f to be

$$\text{grad}(f)=(df)^\#$$

By definition, the gradient of f is the unique section of TM such that

$$(df)_p(w)=g_p(\text{grad}(f)_p,w)$$

for all $p\in M$ and $w\in T_pM$.

Proposition 6.2.2
Let (M,g) be a Riemannian manifold. Let $(U,x^1,\dots,x^n)$ be a chart in M. Let $f:M\to\mathbb{R}$ be a smooth function. Let g be given locally at the chart by

$$g_p=\sum_{1\leq i,j\leq n}g_{i,j}(p)(dx^i)_p\otimes(dx^j)_p$$

Let $(g^{i,j}(p))=(g_{i,j}(p))^{-1}$ be the inverse of the matrix. Then locally at U, the gradient of f at p is given by

$$\text{grad}(f)(p)=\sum_{1\leq i,j\leq n}\frac{\partial f}{\partial x^i}\bigg|_p g^{i,j}(p)\frac{\partial}{\partial x^j}\bigg|_p$$

Example 6.2.3
Let $f:\mathbb{R}^n\to\mathbb{R}$ be a smooth function. Then the gradient of f at $p\in\mathbb{R}^n$ is given by

$$\text{grad}(f)(p)=\sum_{i=1}^n\frac{\partial f}{\partial x_i}\bigg|_p\frac{\partial}{\partial x_i}\bigg|_p$$

Definition 6.2.4 (The Divergence of a Vector Field)
Let (M,g) be a Riemannian manifold. Let $X:M\to TM$ be a smooth vector field. Define the divergence of X to be the unique smooth function $\text{div}(X):M\to\mathbb{R}$ such that for each $p\in M$ and $(U,x^1,\dots,x^n)$ a chart at p, we have

$$(d(X\lrcorner dV))_p=(\text{div}(X)(p))(dV)_p$$

where dV is the local top form given by $(dV)_p=\sqrt{\det(g_{i,j}(p))}(dx^1)_p\wedge\cdots\wedge(dx^n)_p$ and g is given locally at the chart by $g_p=\sum_{1\leq i,j\leq n}g_{i,j}(p)(dx^i)_p\otimes(dx^j)_p$.

Proposition 6.2.5
Let (M,g) be a Riemannian manifold. Let $(U,x^1,\dots,x^n)$ be a chart in M. Let $X:M\to TM$ be a smooth vector field. Suppose that g and X are given locally at the chart by

$$g_p=\sum_{1\leq i,j\leq n}g_{i,j}(p)(dx^i)_p\otimes(dx^j)_p\quad\text{and}\quad X(p)=\sum_{i=1}^n a_i\frac{\partial}{\partial x^i}\bigg|_p$$

respectively. Then locally at U, the divergence of X at $p\in U$ is given by

$$\text{div}(X)(p)=\frac{1}{\sqrt{\det(g_{i,j}(p))}}\sum_{k=1}^n\frac{\partial(a_k\circ\sqrt{\det(g_{i,j})})}{\partial x^k}\bigg|_p$$

Example 6.2.6
Let $X:\mathbb{R}^n\to\mathbb{R}^n$ be a smooth vector field whose components are given by $X_i:\mathbb{R}^n\to\mathbb{R}$. Then the divergence of X at $p\in\mathbb{R}^n$ is given by

$$\text{div}(X)(p)=\sum_{i=1}^n\frac{\partial X_i}{\partial x_i}\bigg|_p$$

Definition 6.2.7 (The Laplacian of a Function)
Let (M,g) be a Riemannian manifold. Let $f:M\to\mathbb{R}$ be a smooth function. Define the Laplacian of f to be the function $\Delta f:M\to\mathbb{R}$ given by

$$\Delta f=\text{div}(\text{grad}(f))$$

Proposition 6.2.8
Let (M,g) be a Riemannian manifold. Let $f:M\to\mathbb{R}$ be a smooth function. Let $(U,x^1,\dots,x^n)$ be a chart in M. Let g be given locally at the chart by

$$g_p=\sum_{1\leq i,j\leq n}g_{i,j}(p)(dx^i)_p\otimes(dx^j)_p$$

Let $(g^{i,j}(p))=(g_{i,j}(p))^{-1}$ be the inverse of the matrix. Then locally at U, the laplcian of f at p is given by

$$\Delta(f)(p)=\frac{1}{\sqrt{\det(g_{i,j}(p))}}\sum_{1\leq k,l\leq n}\frac{\partial(g^{k,l}\circ\sqrt{\det(g_{i,j})}\circ\frac{\partial f}{\partial x^l})}{\partial x^k}\bigg|_p$$

Example 6.2.9
Let $f : \mathbb{R}^n \to \mathbb{R}$ be a smooth function. Then the laplacian of f at $p \in \mathbb{R}^n$ is given by

$$\Delta(f)(p) = \sum_{i=1}^{n} \frac{\partial^2 f}{\partial x_i^2}\bigg|_p$$

6.3 Curl and The Case of Dimension 3

Definition 6.3.1 (The β-Operator)
Let (M, g) be an oriented Riemannian manifold of dimension 3. Define the β-operator to be the map $\beta : TM \to \bigwedge_{i=1}^{2}(T^*M)$ by

$$\beta(X) = X \lrcorner dV$$

Lemma 6.3.2
Let (M, g) be an oriented Riemannian manifold of dimension 3. Then β is a smooth bundle isomorphism.

Definition 6.3.3 (The Curl of a Vector Field)
Let (M, g) be an oriented Riemannian manifold of dimension 3. Let $X : M \to TM$ be a smooth vector field. Define the curl of X to be the vector field $\text{curl}(X) : M \to TM$ given by

$$\text{curl}(X) = \beta^{-1}(d(X^\flat))$$

Proposition 6.3.4
Let (M, g) be an oriented Riemannian manifold of dimension 3. Then the following diagram is commutative:

$$\begin{tikzcd}
\mathcal{C}^\infty(M) \arrow[r, "\text{grad}"] \arrow[d, "\text{id}"'] & \mathfrak{X}(M) \arrow[r, "\text{curl}"] \arrow[d, "\flat"'] & \mathfrak{X}(M) \arrow[r, "\text{div}"] \arrow[d, "\beta"] & \mathcal{C}^\infty(M) \arrow[d, "\flat"] \\
\Omega^0(M) \arrow[r, "d"'] & \Omega^1(M) \arrow[r, "d"'] & \Omega^2(M) \arrow[r, "d"'] & \Omega^3(M)
\end{tikzcd}$$

6.4 Calculus on Manifolds

Let M be an oriented smooth manifold of dimension n. let $p \in \partial M$. Then $T_p(\partial M)$ is an $(n-1)$-dimensional vector subspace of $T_p(M)$. Recall that a boundary chart of M at $p \in M$ is a chart (U, ϕ) such that the following are true.

- $\phi(U) \subseteq \{(x_1, \dots, x_n) \in \mathbb{R}^n \mid x_n \geq 0\}$.
- $\phi^{-1}(\{(x_1, \dots, x_n) \in \mathbb{R}^n \mid x_n = 0\}) \subseteq \partial M$.
- $\phi(p) = 0$.

Definition 6.4.1 (The Unit Normal Vector Field)
Let (M, g) be an oriented Riemannian manifold with boundary. Define the outward point unit normal vector field to be the unique smooth section $\nu : \partial M \hookrightarrow M \to TM$ such that the following are true.

- $g_p(\nu(p), w) = 0$ for all $w \in T_pM$ and all $p \in M$.
- $g_p(\nu(p), \nu(p)) = 1$ for all $p \in M$.
- For any $p \in M$ and any boundary chart (U, ϕ), the nth coordinate of $(\phi_*)(\nu_p)$ is strictly negative.

Lemma 6.4.2
Let (M, g) be an oriented Riemannian manifold of dimension n with boundary. Let (U, ϕ) be the

boundary chart of M. Then locally at $U \cap \partial M$, the unit normal vector field is given by

$$\nu = \frac{1}{\sqrt{g^{nn}}} \sum_{j=1}^{n} g^{jn} \frac{\partial}{\partial x^j}$$

Lemma 6.4.3
Let (M, g) be an orientable Riemannian manifold with boundary. Then we have

$$dS = \iota^*(\nu^\flat \lrcorner dV)$$

Theorem 6.4.4 (The Gradient Theorem)
Let (M, g) be a Riemannian manifold. Let $\gamma : [0, 1] \to M$ be a smooth curve. Let $f \in \mathcal{C}^\infty(M)$ be a smooth function. Then we have

$$\int_\gamma df = f(\gamma(1)) - f(\gamma(0))$$

Theorem 6.4.5 (The Divergence Theorem)
Let (M, g) be a compact oriented Riemannian manifold with boundary. Let $\iota : \partial M \hookrightarrow M$ be the inclusion map. Let $X \in \mathfrak{X}(M)$. Then we have

$$\int_M \operatorname{div}(X)\, dV_g = \int_{\partial M} g(X, \nu)\, dS_g$$

7 The Hodge Star Operator

7.1 The Hodge Star Operator

Definition 7.1.1 (The Hodge Star Operator)
Let (M,g) be an orientable Riemannian manifold. Let $n = \dim(M)$.

- Let $p \in M$. Define $*_p : \Lambda^k T_p^*(M) \to \Lambda^{n-k} T_p^*(M)$ to be the unique $\mathbb{C}$-linear isomorphism such that
$$\alpha \wedge (*_p \beta) = g_p(\alpha, \beta)(dV)_p$$
where $\alpha, \beta \in \Lambda^k T_p^*(M)$ and Vol is the volume form of M associated to g, and $\langle -, - \rangle_p$ is the induced metric on $\Lambda^k(T_p^*(M))$.
- Define the Hodge star operator of M to be the map
$$* \in \Gamma(M, \mathrm{Hom}(\Omega^k(M), \Omega^{n-k}(M)))$$
given by $p \mapsto *_p$.

Lemma 7.1.2
Let (M,g) be an orientable Riemannian manifold. Then the following are true regrading the Hodge star operator.

- $*f = f dV$ for any smooth function $f \in \mathcal{C}^\infty(M)$.
- $**\omega = (-1)^{k(n-k)}\omega$ for any smooth k-form $\omega \in \Omega^k(M)$.

Lemma 7.1.3
Let (M,g) be an orientable Riemannian manifold. Let $X : M \to TM$ be a smooth vector field. Then the following are true.

- $X \lrcorner dV = *(X^\flat)$.
- $\mathrm{div}(X) = (* \circ d \circ *)(X^\flat)$

Lemma 7.1.4
Let (M,g) be an orientable Riemannian manifold of dimension 3. Let $X : M \to TM$ be a smooth vector field. Then we have
$$\mathrm{curl}(X) = (* \circ d(X^\flat))^\#$$

7.2 The L^2-Norm

Definition 7.2.1 (The L^2-Norm)
Let (M,g) be an orientable Riemannian manifold. Define the L^2-norm on M to be the map $\langle -, - \rangle_{L^2} : \Omega^k(M) \times \Omega^k(M) \to \mathbb{R}$ by the formula
$$\langle \alpha, \beta \rangle_{L^2} = \int_M \langle \alpha, \beta \rangle_g dV$$
where $\langle -, - \rangle_g$ is the induce inner product on k-forms.

Lemma 7.2.2
Let (M,g) be an orientable Riemannian manifold. Then $\langle -, - \rangle_{L^2}$ defines an inner product on the compactly supported smooth k-forms on M for any $k \in \mathbb{N}$.

Lemma 7.2.3
Let (M,g) be an orientable Riemannian manifold. Let $\alpha, \beta \in \Omega^k(M)$ be compactly supported smooth k-forms. Then we have
$$\langle \alpha, \beta \rangle_{L^2} = \int_M \alpha \wedge *\beta$$

7.3 The Adjoint Differential Operator

Definition 7.3.1 (The Adjoint Differential Operator)
Let (M, g) be an orientable Riemannian manifold. Define the operator $d^* : \Omega^k(M) \to \Omega^{k-1}(M)$ by

$$d^*(\omega) = (-1)^{n(k+1)+1}(* \circ d \circ *)(\omega)$$

for $\omega \in \Omega^k(M)$. Also define $d^*(\omega) = 0$ for $\omega \in \Omega^0(M)$ by convention.

Lemma 7.3.2
Let (M, g) be an orientable Riemannian manifold. Then we have $d^* \circ d^* = 0$.

Proposition 7.3.3
Let (M, g) be a compact orientable Riemannian manifold. Then we have

$$\langle d^*\omega, \eta \rangle_{L^2} = \langle \omega, d\eta \rangle_{L^2}$$

for any $\omega \in \Omega^{k+1}(M)$ and $\eta \in \Omega^k(M)$.

Definition 7.3.4 (The Laplace-Beltrami Operator)
Let M be an orientable Riemannian manifold. Define the Laplace-Beltrami operator on M to be the map $\Delta : \Omega^k(M) \to \Omega^k(M)$ given by

$$\Delta = d^* \circ d + d \circ d^*$$

Lemma 7.3.5
Let M be an orientable Riemannian manifold. Then the following are true regarding the Laplace-Beltrami operator.
- $* \circ \Delta = \Delta \circ *$.
- $\langle \Delta\alpha, \beta \rangle_{L^2} = \langle \alpha, \Delta\beta \rangle_{L^2}$ for any compactly supported differential forms $\alpha, \beta \in \Omega^k(M)$.

7.4 The Hodge Theorem for Riemannian Manifolds

Definition 7.4.1 (Harmonic Forms)
Let M be an orientable Riemannian manifold. Let $\alpha \in \Omega^k(M)$. We say that α is harmonic if $\alpha \in \ker(\Delta)$.

Lemma 7.4.2
Let M be an orientable Riemannian manifold. Let $\alpha \in \Omega^k(M)$. Then α is harmonic if and only if $d(\alpha) = 0$ and $d^*(\alpha) = 0$.

Definition 7.4.3 (The Space of Harmonic Forms)
Let M be an orientable Riemannian manifold. Define the space of harmonic k-forms to be

$$\mathcal{H}^k(M) = \ker(\Delta)$$

Theorem 7.4.4 (Hodge Theorem)
Let M be an compact orientable Riemannian manifold. Then the following are true.
- $\mathcal{H}^k$ is finite dimensional for all $k \in \mathbb{N}$.
- There is a decomposition
$$\Omega^k(M) \cong \mathcal{H}^k(M) \oplus \operatorname{im}(\Delta)$$

Corollary 7.4.5

> Let M be an orientable Riemannian manifold. Then the quotient map induces an isomorphism
> $$\mathcal{H}^k(M) \cong H^k_{\text{dR}}(M)$$
> for all $k \in \mathbb{N}$.

8 Holonomy

8.1 Holonomy Groups

Definition 8.1.1 (Holonomy Groups)
Let (M,g) be a Riemaniann manifold. Let $p:E\to M$ be a smooth vector bundle. Let ∇ be a connection on E. Let $x\in M$. Define the holonomy group of E at x with respect to ∇ to be

$$\text{Hol}_x(\nabla)=\{P_\gamma\mid\gamma(0)=\gamma(1)=x\}\subseteq\text{GL}(E_x)$$

Lemma 8.1.2
Let (M,g) be a Riemaniann manifold. Let $p:E\to M$ be a smooth vector bundle. Let ∇ be a connection on E. Let $x\in M$. Then $\text{Hol}_x(\nabla)$ is a Lie subgroup of $\text{GL}(E_x)$.

Proposition 8.1.3
Let (M,g) be a Riemaniann manifold. Let $p:E\to M$ be a smooth vector bundle. Let ∇ be a connection on E. Then the following are true.

- Let $\gamma:[0,1]\to M$ be a path. Then we have

$$\text{Hol}_{\gamma(1)}(\nabla)=P_\gamma\text{Hol}_{\gamma(0)}(\nabla)P_\gamma^{-1}$$

- If M is simply connected, then $\text{Hol}_x(\nabla)$ is connected for all $x\in M$.

So for simply connected spaces, the holonomy groups are independent of base point up to isomorphism. But the specific embedding into $\text{GL}(n,F)$ does depend on the base point.

Definition 8.1.4
Let M be a simply connected Riemannian manifold. Let $p:E\to M$ be a smooth vector bundle over $\mathbb{R}$. Let ∇ be a connection on E. Define the holonomy group of E with respect to ∇ to be

$$\text{Hol}(\nabla)=\text{Hol}_x(\nabla)$$

for any base point $x\in M$.

Proposition 8.1.5
Let (M,g) be a Riemaniann manifold. Let $p\in M$. Let $\phi:T_pM\to\mathbb{R}^n$ be an $\mathbb{R}$-linear isomorphism. Then we have

$$\text{GL}(\phi)(\text{Hol}_p(\nabla^g))\leq O(n)$$

Proposition 8.1.6
Let (M,g) and (N,h) be Riemannian manifolds. Let $p\in M$ and $q\in N$. Then we have

$$\text{Hol}_{(p,q)}(\nabla^{g\times h})=\text{Hol}_p(\nabla^g)\times\text{Hol}_p(\nabla^h)$$

Proposition 8.1.7
Let (M,g) be a Riemannian manifold. Let $p\in M$. If M is irreducible, then the representation $\text{Hol}_p(\nabla^g)\to\text{GL}(T_pM)$ is irreducible.

Theorem 8.1.8 (Berger)
Let (M,g) be a Riemannian manifold of dimension n. Suppose that M is simply connected, irreducible and not symmetric. Let $p\in M$. Then exactly one of the following cases hold.

- $\text{Hol}_p(\nabla^g)\cong SO(n)$.
- $n=2m$ for $m\geq 2$ and $\text{Hol}_p(\nabla^g)\cong U(m)\leq SO(n)$.
- $n=2m$ for $m\geq 2$ and $\text{Hol}_p(\nabla^g)\cong SU(m)\leq SO(n)$.
- $n=4m$ for $m\geq 2$ and $\text{Hol}_p(\nabla^g)\cong SP(m)\leq SO(n)$.

- $n = 4m$ for $m \geq 2$ and $\text{Hol}_p(\nabla^g) \cong SP(m) \times SP(1) \leq SO(n)$.
- $n = 7$ and $\text{Hol}_p(\nabla^g) \cong G_2 \leq SO(7)$.
- $n = 8$ and $\text{Hol}_p(\nabla^g) \cong \text{Spin}(7) \leq SO(8)$.